Independent Component Analysis.

Principal Component Analysis is everywhere and it is easy to describe; Independent Component Analysis (ICA) (Comon, 1994; Hyvärinen and Oja, 2000) is often more useful but difficult to explain, or, rather, the need for ICA is easy enough to explain but the problem itself is a difficult one because there isn't really a good, well defined, algorithm for finding the independent components. Like PCA, it is useful to think a bit about it and to learn a bit about what it does, but our treatment is going to be a bit vague. The real message is to think about how things are represented, how changing how data is encoded as a vector, or set of activations over nodes, can disentangle its structure.

The motivation is straightforward and illustrated in Fig. 1. It is common for the distribution of the data to suit PCA. This is what happens if the distribution is a multi-variate Gaussian and for all sorts of reasons this is a common distribution for data; even when the data isn't precisely a multivariate Gaussian, it looks a bit like one. However, it isn't the only possibility. Imagine looking at recipes for example and measuring how much ginger and garlic they contain. If the recipe is Italian the amount of garlic will vary but there will be hardly any ginger; for Indian recipes there will be garlic and ginger and, in fact, they will be proportional, the more garlic, the more ginger; indeed they will have roughly the same amount by weight, so some dishes will have lots of garlic and some little, but those same dishes will also have a lot of ginger or a little. This is the situation we are cartooning in the ICA panel of Fig. 1: the x-axis represents the amount of garlic, the y-axis is the amount of ginger and the dots are recipes, the two red lines pick out the two different cuisines.

This is often when ICA is useful, there is a choice, here, what sort of cuisine, and then a continuous process whose details depend on the discrete choice, here, the amount of ginger and garlic. In this case the axes, horizontal for Italian food, along $x = y$ for Indian, represent the *independent components* in recipe space.

fastICA

I mentioned that we weren't going to look at how to do ICA in any detail. There are basically two main algorithms, *infomax* (Bell and Sejnowski, 1995) and *fastICA* (Hyvärinen and Oja, 1997; Hyvärinen, 1999). The infomax

algorithm is extremely elegant, using ideas from information theory in a tour-de-force of cunning computation to produce a well-motivated approach to finding independent components; applied to patches of natural images it finds edge detectors like those in the visual cortex (Bell and Sejnowski, 1997). However, fastICA is the more popular algorithm, partly because it is fast, partly because the team who developed it put a lot of effort into producing nice, optimized, robust, implementations in lots of languages and finally, I think it helps that the overall reason the algorithm works is less elegant than infomax but easier to explain!

The idea behind fastICA is that mixtures of random processes tend to be Gaussian. There is, in fact, a theorem called the *Central Limit Theorem* that tells us, in a precise way and in precise circumstances, that there is a sense in which

$$S_n = \frac{X_1 + X_2 + \ldots + X_n - n\mu}{\sigma\sqrt{n}} \tag{1}$$

tends towards a random variable with a Gaussian distribution, mean zero and variance one, as n becomes large, when the X_i are independent identically distributed random variables with mean μ and standard deviation σ. The theorem gives precise meanings for 'a sense in which', 'tends towards' and 'becomes large'. However, the precise details of the Central Limit Theorem don't concern us, what is important is a vague lesson we can learn from it: adding random things up usually gives us something that is more like a Gaussian than the things themselves.

Carrying on with our recipe analogy wouldn't be impossible but it might be a bit confusing! Let's consider a more classic example of when ICA is useful and, in fact, one of the main places it is used. This is the *cocktail party problem*. Imagine you are listening in to two people, you and these people might be at a cocktail party, this sounds unlikely in real life, do people still have cocktail parties? But this is just a story. The two people are in different positions and so the sound of their voices is combined when it enters your ear as some linear combination of each, the linear weights depend on how far from your ear they are. Each voice changes amplitude independently, the ear hears a combination of the two. In the way this story is told, you actually have access to two different linear combinations, one for each ear. So, there are two independent sources, each of the two people and you have access to two different linear combinations, one for each ear. The job of ICA is to allow you to separate the two combined signals into two different signals, one for each speaker.

This is completely unrealistic, we are very good at solving the cocktail party problem but we don't solve it by relying on the small difference in amplitude in the signals entering each ear, we use a whole set of techniques and it probably helps that our ears are such an odd shape, but this is just a story and, as mentioned, there are versions of the cocktail party problem where we do have a set of observed signals corresponding to different weighted sums and where we do use fastICA.

So, let's imagine we are in two dimensions and there are two random variables, X_1 and X_2. These are independent, potentially different processes, the two voices in our example, and for this to work they aren't Gaussian. Now let's mix them:

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} = M \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \tag{2}$$

where Y_1 and Y_2 are the values we actually observe, the sounds in our two ears, and M is a mixing matrix, a 2×2 matrix. This means $Y_1 = m_{11}X_1 + m_{12}X_2$ and $Y_2 = m_{21}X_1 + m_{22}X_2$. Thus, the things we observe are more Gaussian than the original variables. The idea is to find an unmixing matrix W so

$$\begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} = W \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} = WM \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \tag{3}$$

Now if $W = M^{-1}$ then $Z_i = X_i$ and we will have found the original unmixed variables. Now the clever thing is that the closer W is to being M^{-1} the less Gaussian we might expect Z_1 and Z_2 to be, the closer each is to being just one of the X_i the less it is a mixture. The idea behind fastICA is to measure how Gaussian the Z_i are and then vary the coefficients of W to minimize whatever measure of "Gaussian-ness" is being used.

A language example

We are going to look at a slightly complicated example, but I think a fun one. It is complicated because it introduces another technology called *word2vec*. The purpose of word2vec is to map words to vectors. The original "skip-gram" word2vec approach (Mikolov et al., 2013a,b) was surprising when it was introduced, it arrived soon after AlexNet (Krizhevsky et al., 2012) and was part of the revival of interest in neural networks. It is very different to AlexNet though since it used a surprisingly shallow network.

The idea is simple: take a big list of all the words in your dictionary, a 10,000-dimensional vector for example, remember this was over a decade

ago. Give all the entries in this vector the value zero. Now pick a word in a text and turn the corresponding entry in your vector to one. Now map your big vector to a smaller continuous vector, say 300-dimensional which we will call the latent space with the activation pattern of that vector for your word just some vector in the latent space $\mathbf{v}$[word] and then map that vector to another big vector corresponding to all words. This involves two big matrices, a 10000×300 matrix mapping the input to the latent space and a 300×10000 matrix to map to the output. Finally map the output vector values to probabilities and train the two big matrices on an objective based on the idea that these probabilities tell us which words are near in the text to the input word. Near here could mean "within eight words either side" or "in the same sentence". After we've done all this the latent space vectors, the $\mathbf{v}$[word]s, are the things we wanted, vectors representing words.

The thing that makes this useful is that it appears to produce vectors that represent aspects of word meaning. Again, this is less remarkable now; we have other more sophisticated transformer based approaches for representing the meanings of words and, indeed, this is one of the things that is happening in the layers of the transformers used in large language models. Here, we are just introducing it as an example of how ICA can pull out the structure of data and it is being used as an example precisely to familiarize you with the way representation works, the way neural networks map things from one vector to another in ways that represent and organize the properties of the things in question. Anyway, in the case of word2vec the original papers showed off a sort of "Euclideanization" of the words, relationships like (Mikolov et al., 2013a, Table 8):

$$\mathbf{v}(\text{Mozart}) + [\mathbf{v}(\text{scientist}) - \mathbf{v}(\text{Einstein})] \approx \mathbf{v}(\text{violinist}) \tag{4}$$

These examples were often highly cherry-picked, but it was still cool at the time!

We won't use skipgram. It comes with a nice explanation but better approaches were found afterwards; instead we use a fairly standard set called *GloVe* (Pennington et al., 2014); GloVe actually finds similar vectors to skipgram (Levy and Goldberg, 2014), 100-dimensional in the version we use, we won't go into the details. Suffice to say, GloVe maps words to vectors and the vectors are easily downloaded from the internet. We are going to do ICA on GloVe, following Yamagiwa et al. (2023), who showed that the independent components of word embeddings are interpretable and, remarkably, much the same across different languages and models.

The idea is to take the GloVe vectors for the twenty thousand or so most common words, leaving out the hundred very commonest ones, words like 'the' and 'of', which turn up everywhere and so tell us very little, and to treat each word's vector as a data point, exactly like the recipes in Fig. 1, except that now there are 100 dimensions rather than two. We run fastICA on these points, asking for 100 components, so that each word gets a new set of coordinates, one for each independent component. To see what a component is 'about' we just list the words with the largest values along it. If we do the same thing with PCA the lists are a bit of a jumble: a few of the principal components are coherent, but most mix unrelated things, so one, for example, has Ingushetia, the Sensex stock index, the PKK and landslides. The independent components are much cleaner, most of them pick out a single category you would recognise, and if we plot the words against two of them, as in Fig. 2, we see the same two-armed picture as the recipes, with food words along one axis, music words along the other and almost nothing in between: most words are mostly about one thing. Here are five of the components, with the words that score highest on each:

- **cooking:** teaspoon, tablespoons, sauce, garlic, chopped, onion, butter
- **music:** albums, vocals, songs, band, guitar, jazz
- **stock markets:** Topix, DAX, KOSPI, FTSE, Nikkei, Nasdaq
- **computing:** desktop, Macintosh, Android, USB, software, server
- **injuries:** spinal, ligament, knee, ankle, tendon, fracture

Not all the components are topics like these, some are about grammar or style rather than subject matter, there is one, for example, made up of intensifiers, incredibly, utterly, extremely, enormously, and another of verbs ending in '-ing'; and a few are junk. Nonetheless, it is striking that a method which knows nothing about language, which only looks for directions along which the data is as un-Gaussian as possible, pulls out so much of the structure of meaning.

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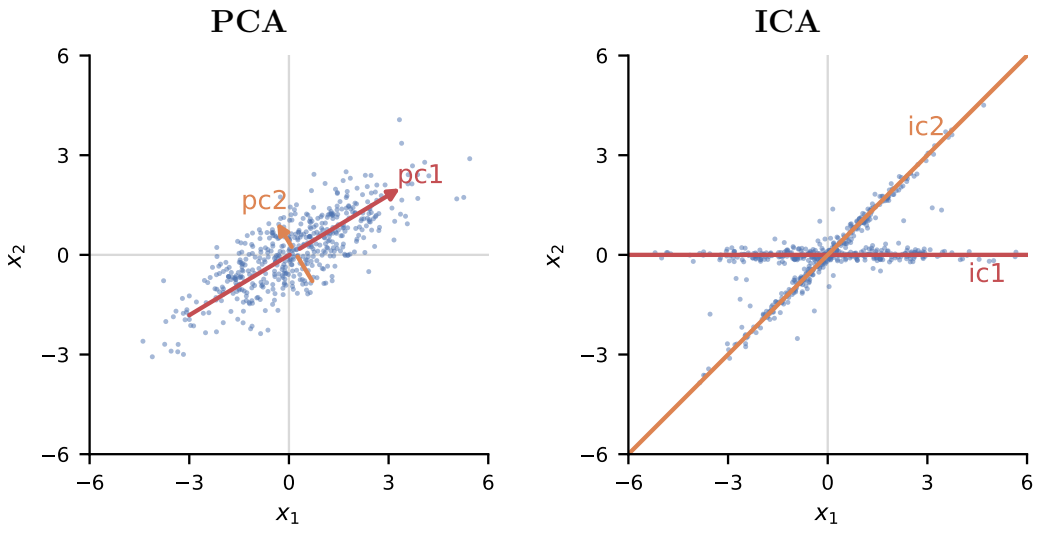


Figure 1: Left: PCA finds orthogonal axes ordered by variance. Right: ICA finds the (here non-orthogonal) directions along which the data are independent.

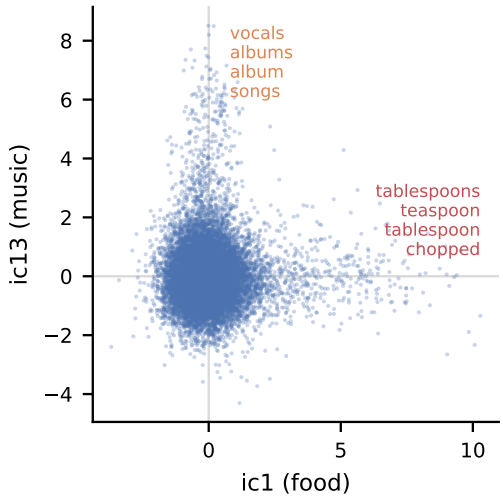


Figure 2: GloVe word vectors plotted against two of the independent components found by fastICA. Most words sit near the origin; words about cooking lie along the horizontal axis and words about music along the vertical one, with very few words scoring highly on both.

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